Internship Handover: Project Involvement Report and Key Contributions (Rutvika Ashtekar)

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Prepared by Rutvika Ashtekar, Cybersecurity Lead, for the EduRankAI internship handover. Over the internship I worked on three cybersecurity streams: the EduRankAI programme Weeks 1 to 3, SAARTHI Phases 1 to 3, and VESPER-SEC-01 Weeks 4 to 7. My internship concludes on Oct 28, 2026, and this report hands the work over.

Project involvement at a glance

Stream	Period	My role	Status at handover
EduRankAI Week 1: CyberOS Firewall Engine	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 5 of 5 tests passed
EduRankAI Week 2: Security Assessment	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 16-risk register
EduRankAI Week 3: Security Posture Analyzer	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 19 of 19 tests passed
SAARTHI SEC-01 Phase 1: Renode AES-GCM and ECDH benchmarks, STRIDE threat model	Before Phase 2	Cybersecurity Lead (Aerospace Security)	Complete; not re-verified in Phase 2 or 3
SAARTHI SEC-01 Phase 2: latency benchmarks, V2X delay model, threat assumptions	14 Oct 2026	Cybersecurity Lead (Aerospace Security)	Delivered; review by Mohan: no changes requested

Stream	Period	My role	Status at handover
SAARTHI SEC-01 Phase 3: ECDSA, Cortex-M4 counts, end-to-end delay model, Zero-Trust spec, fault tests	24 Oct 2026	Cybersecurity Lead (Aerospace Security)	Prepared; technical review and executive sign-off pending
VESPER-SEC-01 Weeks 4 to 7: crypto benchmarking under emulation (Nano-Bee)	Weeks 4 to 7	Author, VISVAMBHARA cybersecurity (Option A)	Weeks 4 to 6 complete; Week 7 implemented, validation pending

Key contributions

EduRankAI, Weeks 1 to 3 (build, assess, automate). I built a firewall engine, assessed it as a security target, then automated the assessment.

- Week 1, CyberOS Firewall Engine: Python modules for packet sniffing, a rule engine, connection tracking, alerts (port-scan and burst heuristics), logging, an API and CLI, and a heuristic risk-scoring assistant. I also ran a Linux firewall lab with nftables, iptables, tcpdump, port scans and a performance baseline.
- Week 2, Security Assessment: asset inventory, data flow, trust boundaries, attack surface, STRIDE-style threat model with MITRE ATT&CK and NIST CSF mapping, vulnerability assessment, a 16-risk register, security architecture and the Week 3 plan.
- Week 3, Security Posture Analyzer: firewall-policy parser, rule-risk analyzer, posture scorer, pipeline, sample scan output, tests, demo recording and presentation.

Recognition after Weeks 1 to 3. The EduRankAI 3-week Research Reality and Originality Assessment (8 September 2026) awarded me Performer Honours for Exceptional Cybersecurity Engineering Performance. It rated my evidence, execution and three-week progression very strong, and placed me on the Viswambhara projects, primary SUDARSHAN and secondary VISBEE. The assessment is in the Drive folder 04_EduRankAI_Weeks_1-3.

SAARTHI SEC-01, Phases 1 to 3 (V2X security for the roadside unit). I owned the cryptography and threat analysis for the ARU.

- Phase 1: STRIDE threat model (threats V2X-S-01 to V2X-E-01, trust boundaries TB-01 to TB-03, security requirements) and Cortex-M4 functional benchmarks of AES-128-GCM and ECDH P-256 in Renode, with code size and static RAM.
- Phase 2: C and Python latency benchmarks of AES-GCM and ECDH, a V2X delay model at 6 Mbps, and Section 12A of the threat model (trust boundaries TB-04 to TB-07, assumptions A-01 to A-10, seven added threats).
- Phase 3: ECDSA P-256 host benchmark, rebuilt Cortex-M4 firmware with Renode instruction counts, an end-to-end delay model, a Zero-Trust specification (ZT-01 to ZT-15, Section 14) and nine fault-injection tests.
- Cross-team inputs I worked with: Deepayan Das (6 Mbps and 10/20 Hz rates), Rakshita Lakkanna B. (FreeRTOS periods and priorities), Mohan (HSM review), Smrithi (ICD v1.1, cell D6).

VESPER-SEC-01, Weeks 4 to 7 (Nano-Bee secure communications).

- Week 4: scope, Cortex-M4 toolchain, Renode and QEMU setup, NOP sanity check (1,000 NOPs measured as 1,003 cycles).
- Week 5: analytical model, cycle and energy model definitions, feasibility criteria.
- Week 6: ran Ascon v1.2, ChaCha20-Poly1305, Kyber-512 and SPHINCS+-Haraka-128f on a Cortex-M4 in Renode; recorded cycles, stack and code size; 22 of 22 project checks passed; 12 evidence screenshots. I wrote the bare-metal integration for SPHINCS+, which comes from the official sphincs/sphincsplus repository and is not my original work.
- Week 7: extended the Ascon benchmark to four input sizes (16, 64, 256 and 1,024 bytes).

Key results and evidence

Every number carries its label: MEASURED (host or Renode), MODELED, or ESTIMATED. Host timings are not Cortex-M4 timings.

Result	Value	Label	Evidence
Firewall engine tests (Week 1)	5 of 5 passed	MEASURED	Repository tests
Posture analyzer tests (Week 3)	19 of 19 passed	MEASURED	Repository tests, scan_results.json

Result	Value	Label	Evidence
AES-GCM added delay, 64 B at 6 Mbps	38.271 µs, 0.04% of a 100 ms period	MODELED	Phase 2 report
ECDSA P-256 sign / verify on host	263.188 µs / 890.081 µs	MEASURED (host)	results_ecdsa.csv, 3 runs
ECDSA P-256 sign / verify on Cortex-M4	37 to 74 ms / 124 to 248 ms	ESTIMATED from Renode counts	m4_estimates.csv
Per-message AES-GCM compute on Cortex-M4, 64 B	0.05 to 0.11 ms	ESTIMATED	m4_estimates.csv
Fault-injection tests	9 of 9 passed; 3 of 3 negative controls showed the weakness	Software simulation	fault_test_results.csv
Zero-Trust requirements	15 specified; 7 verified in simulation (ZT-01, 02, 04, 05, 09, 14, 15)	Specification	Threat model Section 14
Ascon v1.2, 16 B plaintext	9,606 cycles (Week 6)	MEASURED (Renode DWT)	VESPER Week 6 evidence
VESPER project checks	22 of 22 passed	Logic checks, not cycle-accuracy validation	VESPER Week 6 evidence

Main finding for SAARTHI. AES-GCM per-message protection fits both the 10 Hz and 20 Hz periods on the Cortex-M4. Software ECDSA per message does not (124 to 248 ms for one verify against a 100 ms period), even at the optimistic 1.0 cycles per instruction. The recommended design is ECDH plus AES-GCM sessions for unicast links, and HSM or hardware-assisted ECDSA, or a selective-verification policy, for broadcast safety messages. This is a design recommendation, not a measured result.

Honest status and limitations

Nothing below is claimed as confirmed.

- Cortex-M4 times in SAARTHI Phase 3 are ESTIMATED. Renode counts instructions, not cycles; the 168 MHz clock and 1.0 to 2.0 cycles per instruction are assumptions. Real cycle counts need STM32F4 hardware.
- Fault-injection tests simulate protocol logic in Python. They do not test the HSM hardware, the radio or real flash.
- The 64 B telemetry payload, 40 µs PHY overhead and signature sizes are my assumptions. The firmware uses a non-secure test random generator, for timing only.
- Phase 3 criteria C1 (crypto task priority) and C4 (interference from other tasks) are not verified, because Sensing and Communications execution times are not measured.
- Phase 3 technical review (Adarsh Anurag, Deepayan Das) and executive sign-off (Siddharth Prasad) are pending.
- VESPER Week 7: the 256 B and 1,024 B Ascon readings differed between two runs (38,650 vs 25,684 and 137,998 vs 150,964 cycles). Repeatability and DWT accuracy are unresolved, so no performance or feasibility claim is made. Week 8 validation is not done in the files submitted.
- EduRankAI: the evaluator found the firewall prototype-level, not production-grade, and the TCP state tracking simpler than the report described. I do not claim IDS/IPS work beyond the firewall, assessment and posture analyzer.

Pending work and proposed counterparts

Counterparts are my proposals, taken from who each item already depends on. Sir to confirm or reassign.

Pending task	Why it is open	Proposed counterpart
HSM or hardware ECDSA offload; decide broadcast authentication path (ZT-12)	Software ECDSA does not fit the period	K. Mohan Krishna, with Deepayan Das
Review of HSM items ZT-04, 06, 07, 08, 12; HSM nonce-counter persistence test	Phase 3 review not yet received	K. Mohan Krishna
Crypto-task WCET and jitter under FreeRTOS; close criteria C1 and C4	Sensing and Communications timings unmeasured	Rakshita Lakkanna B.

Pending task	Why it is open	Proposed counterpart
Telemetry payload size and key-rotation interval (ZT-06)	Currently my assumptions	Adarsh Anurag, K. Mohan Krishna
Phase 3 IF-DATA submission format and location	Not yet received	Smrithi
Phase 3 technical review and sign-off	Pending	Adarsh Anurag, Deepayan Das; Siddharth Prasad
Optimized P-256 measurement; DWT cycle counts on real STM32F4	Phase 4 hardware	SAARTHI hardware owner (Sir to name)
Sensor plausibility checks and safe state (ZT-13)	Open requirement	Pranav, Adarsh Anurag
VESPER Week 8: repeat Ascon runs, fix replay script, compare with a published result	Week 7 repeatability unresolved	VESPER counterpart (Sir to name)
EduRankAI firewall: harden TCP state machine, production-verify controls	Prototype-level	EduRankAI counterpart (Sir to name)

Google Drive structure and links

All files are in one Google Drive folder, organised by project as below.

Folder	Contents
01_Project_Involvement_Report	This report as PDF, plus an editable copy
02_SAARTHI_SEC-01	Phase 2 and Phase 3 reports, Phase 3 PDF, README, STRIDE threat model, code, CSV results, Evidence screenshots
03_VESPER-SEC-01	Week 4 to 7 reports, README, source, evidence screenshots

Folder	Contents
04_EduRankAI_Weeks_1-3	CyberOS Firewall Engine, security assessment, posture analyzer, tests, demo, assessment report
05_Links	GitHub and Drive links

Main handover folder (all files in one place): Google Drive, Internship Handover. It contains the SAARTHI, VESPER and EduRankAI Weeks 1 to 3 folders linked below.

- SAARTHI: GitHub repository · project site
- EduRankAI Weeks 1 to 3: GitHub repository · project site
- VESPER-SEC-01 Weeks 4 to 7: GitHub repository · project site
- GitHub profile: rutvikaashtekar1607
- LinkedIn: Rutvika Ashtekar